

21ST Experiment

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import make_circles
from sklearn.metrics import accuracy_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Generate Circular Dataset
X, y = make_circles(
    n_samples=500,
    noise=0.1,
    factor=0.4,
    random_state=42
)

# Build Neural Network
model = Sequential([
    Dense(8, activation='relu', input_shape=(2,)),
    Dense(4, activation='relu'),
    Dense(1, activation='sigmoid')
])

# Compile Model
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

```
# Train Model
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```
history = model.fit(  
    X, y,  
    epochs=50,  
    verbose=0  
)
```

```
# Training Loss Graph
```

```
plt.figure(figsize=(5,3))  
plt.plot(history.history['loss'])  
plt.title("Training Loss")  
plt.xlabel("Epoch")  
plt.ylabel("Loss")  
plt.grid(True)  
plt.show()
```

```
# Prediction
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```
pred = model.predict(X)  
pred_class = (pred > 0.5).astype(int)
```

```
# Accuracy
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```
acc = accuracy_score(y, pred_class)  
print("Accuracy:", round(acc*100,2), "%")
```

```
# Plot Circular Dataset
```

```
plt.figure(figsize=(5,4))  
plt.scatter(X[:,0], X[:,1], c=y)  
plt.title("Circular Data")  
plt.xlabel("Feature 1")
```

```
plt.ylabel("Feature 2")
```

```
plt.show()
```

